

2024



International Centre for Clean Water Improves Water Quality and Public Health in India with AWS IoT

The International Center for Clean Water is driving its mission for sustainable clean water using AWS IoT, monitoring soil and water quality in real time with advanced analytics.

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90GB

of sensor data analyzed

2 minutes

to upc



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Advanced data analysis

to identify contamination and uncover possible causes



Scalability

to support projects across India and other countries

Security

with data protected and accessible only to authorized users

Overview

The [International Centre for Clean Water](#) (ICCW) is on a mission to ensure sustainable clean water for all. To more effectively monitor soil and water quality, the ICCW set out to develop an [AWS Internet of Things](#) (IoT) solution to provide real-time data transmission from sensors.

Working with [AWS Partner CloudThat](#), the ICCW developed the IoT solution based on AWS services, including [Amazon Kinesis Data Streams](#) and [Amazon OpenSearch Service](#). The solution provides near real-time updates on soil and water quality, helping data scientists identify pollutants contributing to elevated illness rates in a village in southern India. This innovative approach enhances the ICCW's ability to ensure clean water access and improve public health outcomes.



The International Centre for Clean Water (ICCW) is dedicated to ensuring access to clean water for everyone. As part of the Indian Institute of Technology Madras (IITM) in Chennai, the ICCW operates seven laboratories dedicated to sustainable clean water technologies and collaborates with governments, industries, and universities globally.

To support these initiatives, the Indian Government allocated INR 100 million (\$1.2 million) to the ICCW for funding clean water projects in India over three years. Mr. Nandakumar E, CEO of the International Centre for Clean Water, says, “We’re blessed with abundant rainfall, but we’re still one of the world's water-stressed countries.” The organization’s vision is to apply disruptive technologies to enhance water security and provide clean water to every household in the country.

The ICCW recognized that cloud-based Internet of Things (IoT) technology could expedite results in clean water projects. By connecting ground sensors that measure soil and water quality to the cloud via an IoT network, data scientists can analyze the data to determine water cleanliness and identify potential contaminants. “Our vision was to develop a cloud IoT solution to identify contamination causes and inform policymaking in India,” says Mr. Nandakumar.

Solution | Analyzing Drinking Water Quality with AWS IoT

The ICCW chose Amazon Web Services (AWS) to develop its IoT solution. AWS IoT provided the cloud infrastructure, data management, analytics services, and security needed from a single provider. India-based AWS Partner CloudThat offered consultancy and development expertise to ensure the AWS IoT infrastructure followed best practices.

CloudThat developed the infrastructure around Amazon Kinesis Data Streams, a serverless streaming service that captures, processes, and stores data streams from water quality sensors at scale via an IoT network. [Amazon Data Firehose](#) with [AWS Lambda](#) extracts, transforms, and loads the sensor data into [Amazon Simple Storage Service](#) (Amazon S3) buckets.

With the scalability of Amazon S3, the ICCW can use the service for both short-term data storage and long-term archival. Amazon S3 automatically encrypts data, helping the ICCW maintain security. To further restrict access, CloudThat added [Amazon Cognito](#) to authenticate and manage users, with [Amazon CloudWatch](#) optimizing resources.

To analyze the sensors’ findings, the ICCW adopted Amazon OpenSearch Service and Kibana,  search and analytics suite. With the solution, data scientists can visualize sensor data in web-based dashboards in close to

real time while using [Amazon Athena](#) for further analysis of the data.

With its IoT infrastructure on AWS, the ICCW can stream at least 90GB of sensor data for all-day analysis. The organization can track changes in soil and water quality, with dashboards updated every two minutes. The streaming data is also integrated with the ICCW's internal applications to extract maximum value from the findings, such as real-time water quality and flow monitoring, building digital twins for rapid response to unusual events, and overall effective water resource management. Additionally, the ICCW has developed websites for water quality monitoring and water availability, as well as a community water app to further enhance water resource management. "Our AWS IoT solution allows us to monitor changes in soil and water quality over time, providing clear insights into potential problems and their probable causes," says Mr. Nandakumar.

Outcome | Guiding Government Policies to Develop Clean Drinking Water Strategies

Soon after the development of the IoT solution on AWS, the ICCW participated in a project to investigate elevated incidences of diabetes, cardiovascular disease, and cancers in a village in southern India. One hypothesis was that villagers were becoming ill due to arsenic leaching into their wells from pesticides and fertilizers used in local farming.

To test this hypothesis, the ICCW placed sensors around the village to analyze soil and water quality year-round. The AWS IoT infrastructure helped collect data during the rainy and dry seasons, and during periods of fertilizer use. "The combination of our sensors and the AWS IoT solution made it possible to monitor changes in water and soil quality as they were happening across the growing year," explains Mr. Nandakumar.

The ICCW also accelerated soil and water analysis using laboratory field kits, helping scientists conduct detailed analyses of soil and water on-site. "We could run tests in the village and then upload the data to AWS using a Bluetooth connection," says Mr. Nandakumar.

With the gathered data, the ICCW is correlating its findings with incidents of illness among villagers. If a connection between the villagers' health and drinking water is proven, the local government will consider the ICCW's conclusions for action. "It's still a work in progress, but we're closer to showing whether there is a relationship between the health of villagers and the cleanliness of their water," says Mr. Nandakumar.

Based on the success of this project, the ICCW is supporting similar investigations across southern India. "It's probable that what's happening in one village is also happening in other villages. And if the findings are similar among different villages, we have harder evidence," Mr. Nandakumar comments. "It's clear that our AWS IoT and sensor solution can help local authorities improve the quality of fundamental needs like water and health."

Learn More

To learn more, visit aws.amazon.com/iot.

About International Centre for Clean Water

The International Centre for Clean Water (ICCW) is a research institute located in India, dedicated to developing sustainable clean water solutions using advanced technology. The ICCW focuses on reducing water contamination



and improving the management of water resources. Its mission is to enhance water security in India and around the world, ensuring access to clean and safe water for all.

AWS Services Used

Amazon Kinesis Data Streams

Amazon Kinesis Data Streams is a serverless streaming data service that makes it easy to capture, process, and store data streams at any scale.

[Learn more »](#)

Amazon Data Firehose

Amazon Data Firehose provides the easiest way to acquire, transform, and deliver data streams within seconds to data lakes, data warehouses, and analytics services.

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Amazon OpenSearch Service

Amazon OpenSearch Service makes it easy for you to perform interactive log analytics, real-time application monitoring, website search, and more.

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Amazon S3

Amazon Simple Storage Service (Amazon S3) is an object storage service offering industry-leading scalability, data availability, security, and performance.

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